

LCA OF ALUMINIUM VS CARBON FIBRE BIKE FRAMES



Aluminium and carbon fiber composites are the most chosen materials for bike frames due to their favorable strength to weight ratios and performance.

Global bicycle market is projected to grow at a Compound Annual Growth Rate (CAGR) of approximately 6.7 % from 2023 to 2028, reaching an additional \$25.54 billion by 2028.

Aluminium

Aluminium bike frame production begins with energy-intensive bauxite mining and refining, using virgin aluminium.



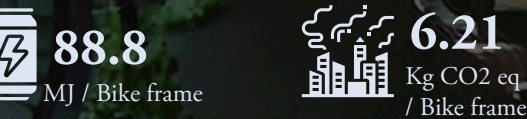
Aluminium bike frame manufacturing involves extrusion, hydroforming, cutting, TIG welding, heat treatment, and painting, requiring 2.6 kg of aluminium per frame due to material wastage.



The use-phase of carbon fiber and aluminium bike frames is identical, involving 10,000 km of human-powered operation with minimal environmental impact before scrapping.



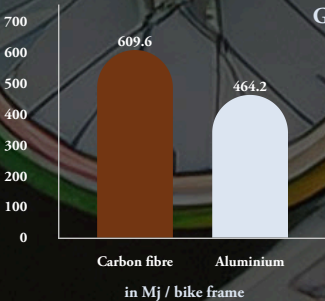
The aluminium bike frame recycling phase involves cutting the frame for handling, with 95% of material recovered, reducing energy use by 95% compared to primary production, and granting credits for avoided raw material extraction.



Energy recovered and Kg CO2 displaced by recycling



Net
Energy
consumed



Carbon fibre

Carbon fiber production involves converting raw materials like ammonia and propylene into polyacrylonitrile (PAN) fibers, which are oxidized and treated, while epoxy resin is formed from petrochemical components and combined with carbon fibers into prepreg sheets for bike frame manufacturing



Carbon fiber bike frame manufacturing involves layering epoxy resin prepreg sheets into precision aluminium molds, autoclave curing for strength, post-processing for refinement, and painting for a high-quality finish.



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End-of-life processes for carbon fiber bike frames include landfill, incineration, mechanical recycling, pyrolysis, fluidized bed recycling, and chemical recycling, with an average of all methods considered for a balanced assessment due to data limitations.



- Landfill widely used option- less energy consumption and Co2 eq emissions, but does not align with CE.
- Pyrolysis and Chemical recycling- Highest impact
- Material recovery of Cf through recycling is very less and has poor quality.
- Technological advancements in recycling methods of CF needed.



Net
CO2 eq
emissions

